LINUX FUNDAMENTALS

Detailed Training Guide | Beginner to Practical Command Line

Concepts, architecture, command examples, expected behavior, security basics, troubleshooting, hands-on labs, interview questions and a suggested training plan.

Audience: Students, new developers, DevOps beginners and IT professionals.

Lab environment: Ubuntu, Debian or Kali Linux terminal. Some package commands differ on other distributions.

Important: Run destructive commands only on disposable practice files. Do not run commands as root unless required.

1. What is Linux?

Linux is the kernel: the core software that manages hardware, processes, memory, devices and system calls. A usable Linux distribution combines the kernel with tools, libraries, package management and applications. Linux is widely used for web servers, containers, cloud infrastructure, embedded systems, development and security testing.

Kernel vs shell vs terminal

Term	Meaning	Example
Kernel	Manages hardware and system resources	Linux kernel
Shell	Interprets typed commands and runs programs	bash, zsh
Terminal	Interface used to interact with a shell	GNOME Terminal
Distribution	Packaged operating system	Ubuntu, Debian, Fedora, Kali

Popular distributions

Ubuntu: beginner-friendly desktop and server distribution. Debian: stability-oriented distribution. Fedora: newer software and development workflows. RHEL: enterprise-focused. Kali: security assessment distribution; it is not required for learning basic Linux.

2. Linux architecture and startup

```
User / Application
        |
Shell / Libraries / System Calls
        |
Linux Kernel
        |
CPU / RAM / Disk / Network / Devices
```

At boot, firmware starts a bootloader (commonly GRUB), which loads the Linux kernel and initial RAM filesystem. The kernel initializes hardware and starts the first userspace process (often systemd). Services then start and the system provides login interfaces.

```
uname -r        # kernel release
uname -a        # detailed kernel/system information
cat /etc/os-release   # distribution details
whoami          # logged-in username
hostnamectl     # host details on systemd systems
```

3. Using the command line

A typical command has a program name, options and arguments: command [options] [arguments]. Linux paths and most command names are case-sensitive. Use Tab for completion and the Up arrow to reuse history.

```
pwd
ls
ls -lah
```

```
cd /tmp
pwd
cd ~
clear
history | tail -n 10
```

Command	Explanation
pwd	Print working directory
ls -lah	Long list; include hidden entries; human-readable sizes
cd ..	Move to parent directory
cd ~	Move to your home directory
man ls	Read the ls manual
command --help	Quick command-specific help (when supported)

4. Linux file system

```
/
|-- home/   user home directories
|-- etc/    system configuration
|-- var/    logs, caches and variable data
|-- usr/    programs, libraries, shared files
|-- tmp/    temporary files
|-- dev/    device nodes
|-- proc/   virtual process/kernel information
|-- sys/    virtual device/kernel information
|-- boot/   boot-related files
`-- root/   root account home
```

An absolute path begins at / (example: /home/student/notes.txt). A relative path begins at the current directory (example: docs/notes.txt). A filename beginning with a dot is hidden from default ls output.

```
mkdir -p ~/LinuxLab/docs
cd ~/LinuxLab
touch docs/notes.txt
ls -la docs
realpath docs/notes.txt
```

5. Creating, viewing, copying and deleting files

```
echo "Linux is useful" > notes.txt
cat notes.txt
printf "second line\n" >> notes.txt
head -n 1 notes.txt
tail -n 1 notes.txt
cp notes.txt backup.txt
mv backup.txt archive.txt
mkdir -p old
mv archive.txt old/
rm old/archive.txt
rmdir old
```

Redirection: > replaces file contents; >> appends. Deletion: rm does not generally use a recycle bin. Avoid rm -rf on unfamiliar paths; check with pwd and ls first.

6. Linux permissions and ownership

Each file has an owner, a group and permissions for owner/group/others. For regular files: read (r) allows reading, write (w) allows modifying, execute (x) allows running. For directories: x allows traversal; r allows listing names; w together with x permits changing directory entries.

```
ls -l notes.txt
# -rw-r--r-- 1 student student ... notes.txt
chmod 640 notes.txt
chmod u+x script.sh
chmod go-rwx secret.txt
id
groups
```

Octal	Owner	Group	Others
755	rwx	r-x	r-x
644	rw-	r--	r--
700	rwx	---	---
600	rw-	---	---

Read=4, write=2, execute=1. For example 750 means 7 (rwx), 5 (r-x), 0 (---). chown changes ownership and normally requires elevated privileges. Use sudo only for commands that genuinely need administrative access.

7. Users, groups and sudo

```
whoami
id
groups
getent passwd "$USER"
sudo -l
# Administrative example (Ubuntu/Debian):
sudo adduser trainee
sudo usermod -aG sudo trainee
```

Do not run user creation on a shared or managed machine without authorization. The root user has extensive privileges. The sudo command runs an authorized command with elevated privileges and can be audited.

8. Processes, jobs and signals

A process is a running program with a process ID (PID). A job is a command managed by the current shell. A signal requests an action from a process; SIGTERM (15) asks for graceful termination, whereas SIGKILL (9) forcibly stops it.

```
ps aux | head
top
```

```
sleep 120 &
jobs
ps -ef | grep sleep
# replace PID with a process ID you own
kill PID
# Ctrl+C interrupts a foreground command
```

Useful concepts: foreground vs background, parent and child processes, exit status, CPU usage and memory usage. echo $? displays the previous command exit status; 0 generally means success.

9. Software installation and updates

On Debian-based distributions such as Ubuntu and Kali, apt downloads packages from configured repositories. Fedora/RHEL commonly use dnf instead.

```
sudo apt update
apt search tree
sudo apt install tree
tree ~/LinuxLab
sudo apt remove tree
apt list --installed | head
```

apt update refreshes package indexes; it does not install all upgrades. apt upgrade upgrades eligible installed packages. Verify repository trust before installing software.

10. Searching and processing text

```
printf "Alice\nBob\nAlice\n" > names.txt
cat names.txt
grep -n "Alice" names.txt
sort names.txt | uniq -c
wc -l names.txt
find ~/LinuxLab -type f -name "*.txt"
cat names.txt | head -n 2
```

A pipe (|) sends one program's standard output into another program's standard input. Useful commands include grep (match lines), cut (select fields), sort (order lines), uniq (adjacent duplicate handling), wc (counts) and find (file discovery).

11. Disk, memory and system monitoring

```
df -h                 # mounted filesystem space
du -sh ~/LinuxLab      # directory usage
free -h               # RAM and swap
lsblk                 # block devices
uptime                # uptime and load average
cat /proc/cpuinfo | head
```

Distinguish total disk capacity from free space, and physical RAM from swap. Load average is not the same as CPU percentage. The /proc filesystem exposes kernel-maintained runtime information.

12. Networking fundamentals

An IP address identifies an interface on a network; a default route identifies where packets are sent outside local subnets; DNS resolves names into addresses. Listening ports identify network services. TCP is connection-oriented; UDP is datagram-based.

```
ip -br addr
ip route
ping -c 4 1.1.1.1
getent hosts example.com
ss -tuln
curl -I https://example.com
# Local HTTP test:
python3 -m http.server 8000
```

Open http://127.0.0.1:8000 from the same machine while the server runs; press Ctrl+C to stop it. A successful ping to an IP but failed hostname lookup can indicate a DNS issue. Some networks block ICMP ping, so ping failure alone does not prove a host is offline.

13. Services and system logs

Many modern distributions use systemd to manage services. Some containers and minimal systems do not run systemd as PID 1.

```
systemctl status ssh
systemctl list-units --type=service --state=running
journalctl -n 30 --no-pager
journalctl -u ssh -n 20 --no-pager
# On some systems service name is sshd instead of ssh
```

Use service logs to diagnose failures, but avoid copying secrets or private data from logs into public forums.

14. Shell scripting basics

A shell script is a file containing commands. A shebang tells the system which interpreter to use. Quote variable expansions to preserve spaces and reduce unexpected word splitting.

```
#!/usr/bin/env bash
set -e
name="Student"
echo "Hello, $name"
for n in 1 2 3; do
  echo "Lab number: $n"
done
if [ -d "$HOME/LinuxLab" ]; then
  echo "Lab directory exists"
fi
```

Save as hello.sh; run bash hello.sh or set executable permission with chmod +x hello.sh and run ./hello.sh. For production scripts, learn input validation, safe quoting, error handling and shellcheck.

15. Environment variables and command discovery

```
echo "$HOME"
echo "$PATH"
which python3
command -v bash
export COURSE_NAME="Linux Fundamentals"
echo "$COURSE_NAME"
printenv | head
```

Environment variables influence program behavior. PATH is the ordered list of directories searched for executable commands. Avoid printing full environments in shared logs because they may contain credentials.

16. Archives and compression

```
tar -czf linuxlab.tar.gz -C "$HOME" LinuxLab
tar -tzf linuxlab.tar.gz | head
mkdir -p ~/restore-test
tar -xzf linuxlab.tar.gz -C ~/restore-test
ls ~/restore-test/LinuxLab
```

tar packages files; gzip compresses them. When extracting untrusted archives, inspect filenames and extract to a controlled directory to avoid overwriting valuable files.

17. Common troubleshooting workflow

Problem	Diagnostic commands	Likely next step
Command not found	command -v NAME; echo "$PATH"	Check spelling or package installation
Permission denied	id; ls -ld PATH; ls -l FILE	Check ownership and permissions
Disk full	df -h; du -sh DIR	Locate large files safely
Network unreachable	ip addr; ip route; ping -c 2 GATEWAY	Check interface, route and firewall
Service not running	systemctl status NAME; journalctl -u NAME	Read error details before changing settings

18. Practical labs with solutions

Lab A - File management

Task: Create a folder, three files, write data, copy a file, rename a file and display results.

```
mkdir -p ~/LinuxLab/assignment
cd ~/LinuxLab/assignment
touch a.txt b.txt c.txt
echo "Hello Linux" > a.txt
cp a.txt backup.txt
mv b.txt renamed.txt
ls -l
cat backup.txt
```

Lab B - Permissions

Task: Make a file readable/writable only by its owner and verify.

```
echo "private" > private.txt
chmod 600 private.txt
ls -l private.txt
# Expected permission prefix: -rw-------
```

Lab C - Search and count

```
printf "error\nok\nerror\nwarning\n" > app.log
grep -n error app.log
grep -c error app.log
# Expected count: 2
```

Lab D - Network and HTTP

```
ip -br addr
ip route
python3 -m http.server 8000
# In another terminal:
curl -I http://127.0.0.1:8000
```

Lab E - Simple backup

```
tar -czf "$HOME/linux-backup.tar.gz" -C "$HOME" LinuxLab
tar -tzf "$HOME/linux-backup.tar.gz" | head
```

19. Interview and viva questions

Q1. What is the difference between Linux and Ubuntu?
Linux is the kernel; Ubuntu is a distribution built around it.

Q2. What does pwd do?
Prints the current working directory.

Q3. What is the difference between absolute and relative paths?
Absolute starts at /; relative is resolved from the current directory.

Q4. What does chmod 755 mean?
Owner rwx; group r-x; others r-x.

Q5. What is sudo?
A mechanism for running authorized commands with elevated privileges.

Q6. Difference between cp and mv?
cp copies; mv moves or renames.

Q7. What is a process ID?
A numeric identifier assigned to a running process.

Q8. What does grep do?
Filters lines matching a pattern.

Q9. What does the pipe operator do?
Connects one command's stdout to another command's stdin.

Q10. How do you inspect listening ports?
Use ss -tuln.

Q11. What is the difference between apt update and apt upgrade?
update refreshes package metadata; upgrade installs eligible package updates.

Q12. What is the root directory?
/ is the top of the Linux filesystem hierarchy.

Q13. What is a shell script?
A file of shell commands executed by a shell interpreter.

Q14. How do you check disk and memory?
df -h for filesystem usage; free -h for memory.

Q15. What is a daemon?
A background service process, often managed by a service manager.

20. Suggested 5-day teaching plan

Day	Concepts	Practical outcome
1	Linux overview, architecture, terminal, navigation	Explore directories and commands
2	Files, permissions, users and groups	Create, secure and manage files
3	Processes, packages, disk and text tools	Monitor system and process logs
4	Networking, services and troubleshooting	Run and diagnose a local HTTP server
5	Shell scripting, backup and review	Complete integrated Linux lab

21. Quick-reference command sheet

Navigation: pwd, ls, cd, realpath. Files: touch, mkdir, cp, mv, rm, cat, head, tail. Security: ls -l, chmod, chown, id, sudo. Processes: ps, top, jobs, kill. Text: grep, find, sort, uniq, wc. System: df, du, free, lsblk, uname. Network: ip, ping, ss, curl. Packages: apt. Services: systemctl, journalctl. Archives: tar.

Further reading: man pages (man bash, man chmod, man find), GNU Coreutils documentation, Ubuntu Server documentation, and Linux Foundation Introduction to Linux.